

COM-480 - Data Visualization Milestone 2

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World Cinema Explorer

The aim of our project is to allow users to discover world cinema through a highly interactive interface combining geographic, temporal, social, and visual lenses.

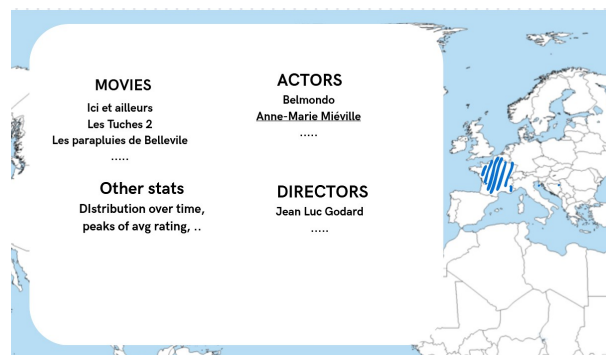
1 Visualizations

1.1 Interactive Choropleth World Map

The map is the primary entry point. Countries are encoded by color saturation, mapping a selected metric (average film rating, total output) onto a sequential color scale rendered on an interactive 3D globe.

Clicking a country triggers a linked detail panel showing the highest-rated films (with poster thumbnails), top actors and directors ranked by frequency or average rating, and genre distribution, average rating, and film count.

Tools: globe.gl (Three.js) + GeoJSON. Course: Maps



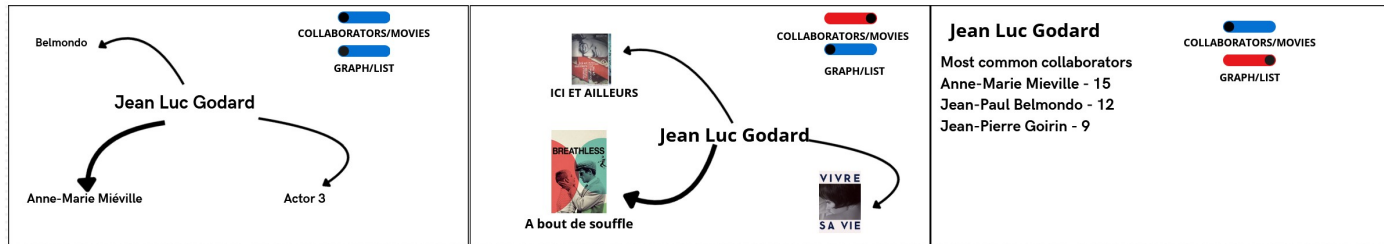
1.2 Collaboration Graph

The network view encodes the social structure of cinema. A force-directed graph is centered on a chosen filmmaker, with edges radiating to all collaborators present in the dataset. Edge weight (thickness + opacity) is proportional to the number of shared films.

A binary toggle switches between two view modes for a given filmmaker:

- **Graph / List** : visualize collaborations as a force-directed network, or as a ranked list of top collaborators with shared film counts
- **Collaborators / Movies** : focus the view on people they worked with, or on the films themselves

Tools: D3-force and Sigma.js. Course: Interactions, Marks & Channels



1.3 Movie Detail Panel

A shared side panel (reachable from both the map and the network) displays a single film entry: poster, director, release year, cast, and rating. It is the semantic bridge between geographic and relational perspectives. As illustrated in the sketch with “Ici et ailleurs” (Godard, 1976), the panel appears on country click or edge click and updates reactively.

Ici et ailleurs



DIRECTED BY Jean Luc Godard,
Anne-Marie Miéville

Released 1976

Genre(s) : Documentary

Primary Language : French

Spoken Languages : Arabic, French, German

Themes : ...

Description : Here and Elsewhere takes its name from the contrasting footage it shows of the fedayeen and of a French family watching television at home...

CREW
Jean-Pierre Gorin
...

CAST
Anne-Marie Miéville
...

2 Extensions and Additional Features

2.1 Time Slider

A slider (range: 1874–2025) filters all views by year or year range. Implemented in Svelte/Vanilla JS; D3 rescales the color domain on each update. This applies semantic zoom to the time axis.
Course: Interactions

2.2 Export & Co-production Flow Lines

Directional arcs connect countries linked by co-productions or significant export relationships. Arc thickness is weighted by film count (same logic as the network edges). An ON/OFF toggle prevents visual overload.

2.3 Extension 1: Period Visual Effects

CSS filter properties are applied globally as the slider crosses era boundaries: grayscale for pre-1950 (black-and-white era), saturate for 1950s–60s Technicolor, grain overlay for the 1970s. The effect is triggered by threshold comparisons on the slider value and reinforces the temporal encoding perceptually.

2.4 Extension 2: Visual Poster Wall

A tab renders clusters of visually similar film posters together. The frontend sorts and lays out the grid in CSS.